

Terraform Provider

Series: AUTOM | **Notebook:** 4 of 8 | **Created:** January 2026 | **Last Updated:** 02/11/2026

The Dynatrace Terraform provider enables infrastructure-as-code management of Dynatrace configurations. It integrates with Terraform's ecosystem for state management, planning, and CI/CD integration.

Table of Contents

1. [Introduction](#)
 2. [Getting Started](#)
 3. [Provider Configuration](#)
 4. [Resource Types](#)
 5. [State Management](#)
 6. [Advanced Patterns](#)
-

Prerequisites

Before starting this notebook, ensure you have:

Requirement	Description
Terraform CLI	Version 1.0+ installed
Authentication	One or more of: API Token (classic), Platform Token , or OAuth Client (see Provider Configuration)
Tenant URL	Your Dynatrace SaaS tenant URL
HCL Knowledge	Basic familiarity with Terraform syntax

Authentication Methods — Three Token Types

The Dynatrace Terraform provider supports three authentication methods. Each covers a different set of resources:

Token Type	Format	Covers	Cannot Cover
API Token (classic)	<code>dt0c01.xxxx</code>	Settings 2.0, Synthetics, SLOs	Gen3 Platform (workflows, documents, segments)
Platform Token	<code>dt0s16.xxxx</code>	Settings 2.0 + Gen3 Platform	Synthetics, SLOs (removed in v1.88.0)
OAuth Client	Client ID + Secret	Gen3 Platform, IAM	Settings 2.0, Synthetics, SLOs (removed in v1.88.0)

Important (v1.88.0): As of Dynatrace Terraform provider **v1.88.0**, OAuth-based authentication (including Platform Tokens acting as OAuth) **can no longer manage synthetic monitors or SLO definitions**. These resources require a classic **API Token** with the appropriate scopes. When both tokens are configured, the provider automatically uses the correct one for each resource.

Recommended setup: Use **Platform Token + API Token** together for full resource coverage. The Platform Token handles Settings 2.0 and Gen3 resources; the API Token handles Synthetics and SLOs.

Learning Objectives

By the end of this notebook, you will:

- Understand the Dynatrace Terraform provider
 - Know how to configure resources in HCL
 - Be able to manage state and handle drift
 - Implement multi-environment deployments
-

1. Introduction

Why Terraform?

Benefit	Description
State Management	Track what exists vs. what's defined
Drift Detection	Identify manual changes
Planning	Preview changes before applying
Ecosystem	Integrate with other Terraform providers
Modules	Reusable configuration packages

Terraform vs Monaco

Aspect	Terraform	Monaco
State file	Required	None
Drift detection	Built-in	Manual
Learning curve	Higher	Lower
Multi-cloud	Yes	Dynatrace only
Dependencies	Explicit	Implicit

2. Getting Started

Installation

Install Terraform:

macOS:

```
brew tap hashicorp/tap
brew install hashicorp/tap/terraform
```

Linux:

```
wget -O- https://apt.releases.hashicorp.com/gpg | sudo gpg --dearmor -o
/usr/share/keyrings/hashicorp-archive-keyring.gpg
echo "deb [signed-by=/usr/share/keyrings/hashicorp-archive-keyring.gpg]
https://apt.releases.hashicorp.com $(lsb_release -cs) main" | sudo tee
/etc/apt/sources.list.d/hashicorp.list
sudo apt update && sudo apt install terraform
```

Verify installation:

```
terraform version
```

3. Provider Configuration

The Dynatrace Terraform provider supports three authentication methods. Which you need depends on the resources you manage.

Method 1: API Token (Classic)

Use for **Settings 2.0** resources and resources that require classic API token auth (synthetic monitors).

API tokens require explicit scopes like `settings.read`, `settings.write`, `ExternalSyntheticIntegration`.

```

terraform {
  required_providers {
    dynatrace = {
      source  = "dynatrace-oss/dynatrace"
      version = "~> 1.0"
    }
  }
}

provider "dynatrace" {
  dt_env_url    = var.dynatrace_url
  dt_api_token = var.dynatrace_token # Classic API token (dt0c01.xxxx)
}

```

Method 2: Platform Token

Platform Tokens (`dt0s16.xxxx`) are a newer token type that bridges Settings 2.0 and Gen3 Platform in a single credential. They work within the assigned user's permissions.

```

provider "dynatrace" {
  # Platform Token handles Settings 2.0 + Gen3 Platform resources
  # Set via DYNATRACE_ENV_URL and DYNATRACE_PLATFORM_TOKEN env vars
}

```

Key env var: `DYNATRACE_HTTP_OAUTH_PREFERENCE=true` tells the provider to use the Platform Token for OAuth-based resources (workflows, documents, segments) in addition to Settings 2.0.

Method 3: OAuth Client Credentials

Required for Automation (Workflows), Document, and Account Management (IAM) resources. The provider exchanges your client ID and secret for short-lived OAuth access tokens automatically.

```

provider "dynatrace" {
  dt_env_url    = var.dynatrace_url
  dt_api_token = var.dynatrace_token # For settings/classic resources

  # OAuth credentials – required for automation, document, and IAM resources
  automation_client_id      = var.automation_client_id
  automation_client_secret = var.automation_client_secret
}

```

Tip: If you only manage automation/document resources, you can omit `dt_api_token` and use OAuth alone.

Method 4: Combined Auth — Full Coverage (Recommended)

For **full resource coverage**, use **Platform Token + API Token** together. The provider automatically uses the correct token for each resource type:

```
# provider.tf — Combined auth for full coverage
provider "dynatrace" {
  # Auth is read from environment variables automatically.
  # The provider uses the correct token for each resource type.
}
```

```
# --- Platform Token (covers Settings 2.0 + Gen3 Platform) ---
export DYNATRACE_ENV_URL="https://abc12345.live.dynatrace.com"
export DYNATRACE_PLATFORM_TOKEN="dt0s16.xxxx.yyyy"
export DYNATRACE_HTTP_OAUTH_PREFERENCE=true

# --- API Token (covers Synthetics — v1.88.0 requirement) ---
export DYNATRACE_API_TOKEN="dt0c01.xxxx.yyyy"
```

Resource	Token Used
Settings 2.0 (auto-tags, management zones, alerting)	Platform Token
Gen3 Platform (workflows, documents, segments)	Platform Token (via OAuth)
SLO definitions (<code>dynatrace_slo_v2</code>)	Platform Token (Settings 2.0 — <code>builtin:monitoring.slo</code>)
Synthetic monitors (<code>dynatrace_http_monitor</code>)	API Token (v1.88.0)

Environment Variables — Complete Reference

Variable	Purpose
DYNATRACE_ENV_URL or DT_ENV_URL	Tenant URL
DYNATRACE_API_TOKEN or DT_API_TOKEN	Classic API token (dt0c01)
DYNATRACE_PLATFORM_TOKEN	Platform token (dt0s16)
DYNATRACE_HTTP_OAUTH_PREFERENCE	Set to true to use Platform Token for Gen3 resources
DT_CLIENT_ID	OAuth client ID (for automation/document/IAM resources)
DT_CLIENT_SECRET	OAuth client secret
DT_ACCOUNT_ID	Account UUID (required for account-level IAM resources)

Which Authentication for Which Resources?

Resource Category	API Token	Platform Token	OAuth Client
Settings 2.0 (management zones, auto-tags, alerting, SLOs)	Yes	Yes	No
Synthetic monitors	Yes (required)	No (v1.88.0)	No (v1.88.0)
Gen3: Workflows, scheduling	No	Yes (with <code>HTTP_OAUTH_PREFERENCE</code>)	Yes
Gen3: Documents (dashboards, notebooks)	No	Yes (with <code>HTTP_OAUTH_PREFERENCE</code>)	Yes
Gen3: Segments	No	Yes (with <code>HTTP_OAUTH_PREFERENCE</code>)	Yes
Account Management (IAM, policies, groups)	No	No	Yes (<code>DT_ACCOUNT_ID</code> also needed)

Creating an OAuth Client

1. Go to **Account Management > Identity & Access Management > OAuth clients**
2. Create a new client with the required scopes:

Scope	Purpose
<code>automation:workflows:read</code>	Read workflow definitions
<code>automation:workflows:write</code>	Create/update workflows
<code>automation:rules:read</code>	Read scheduling rules
<code>automation:rules:write</code>	Create/update scheduling rules
<code>document:documents:read</code>	Read documents
<code>document:documents:write</code>	Create/update documents
<code>document:documents:delete</code>	Delete documents
<code>storage:filter-segments:read</code>	Read segments
<code>storage:filter-segments:write</code>	Write segments
<code>settings:objects:read</code>	Read settings (if IAM policy allows)
<code>settings:objects:write</code>	Write settings (if IAM policy allows)

3. Copy the generated **client ID** and **client secret** immediately — the secret is only shown once
4. Store credentials securely (vault, CI/CD secrets, etc.)

Security: Never commit OAuth client secrets or API tokens to version control. Use environment variables, HashiCorp Vault, AWS Secrets Manager, or your CI/CD platform's secret store.

Note on scope formats: API token scopes use dot notation (`settings.read` , `settings.write`). OAuth scopes and IAM policy actions use colon notation (`settings:objects:read` , `settings:objects:write`). These are different identifiers for the same capability in different auth contexts.

Initialize and Validate

```
# Initialize provider
terraform init

# Validate configuration
terraform validate

# Format code
terraform fmt
```

4. Resource Types

Management Zone

```
resource "dynatrace_management_zone_v2" "production" {
  name = "Production"

  rules {
    type      = "SERVICE"
    enabled   = true

    conditions {
      condition {
        key {
          type      = "STATIC"
          attribute = "SERVICE_TAGS"
        }
        tag {
          operator = "EQUALS"
          negate   = false
          value {
            context = "CONTEXTLESS"
            key     = "environment"
            value   = "production"
          }
        }
      }
    }
  }
}
```

Auto-Tagging Rule

```
resource "dynatrace_autotag_v2" "application" {
  name = "Application"

  rules {
    type      = "SERVICE"
    enabled   = true
    value_format = "{Service:DetectedName}"
    conditions = []
  }
}
```

Alerting Profile

```
resource "dynatrace_alerting" "production_alerts" {
  name          = "Production Alerting"
  management_zone = dynatrace_management_zone_v2.production.id

  rules {
    delay_in_minutes = 0
    severity_level   = "AVAILABILITY"
    tag_filters      = []
  }

  rules {
    delay_in_minutes = 5
    severity_level   = "ERRORS"
    tag_filters      = []
  }

  rules {
    delay_in_minutes = 10
    severity_level   = "PERFORMANCE"
    tag_filters      = []
  }
}
```

SLO

```
resource "dynatrace_slo_v2" "availability" {
  name           = "Production Availability"
  enabled        = true
  evaluation_type = "AGGREGATE"
  evaluation_window = "-1w"
  target_success  = 99.9
  target_warning  = 99.95

  metric_expression =
    "builtin:synthetic.http.availability.location.total:splitBy()"

  filter          = "type(SYNTHETIC_TEST)"
}
```

HTTP Monitor (Synthetic)

```
resource "dynatrace_http_monitor" "homepage" {
  name      = "Homepage Check"
  enabled   = true
  frequency = 5

  locations = ["GEOLLOCATION-1234567890ABCDEF"]

  anomaly_detection {
    loading_time_thresholds {
      enabled = true
    }
    outage_handling {
      global_outage = true
      local_outage  = false
    }
  }

  script {
    request {
      description = "Homepage"
      method      = "GET"
      url          = "https://example.com"

      validation {
        rule {
          type = "httpStatusesList"
          value = ">=400"
          pass_if_found = false
        }
      }
    }
  }
}
```

Gen3 Platform Resources

Gen3 resources require **Platform Token** (with `DYNATRACE_HTTP_OAUTH_PREFERENCE=true`) or **OAuth Client Credentials**.

Automation Workflow

```
resource "dynatrace_automation_workflow" "problem_email" {
  title      = "Problem Email Notification"
  description = "Sends an email when a Davis problem opens"
  private    = false

  trigger {
    event {
      active = true
      config {
        davis_problem {
          categories {
            error = true
          }
          entity_tags = {
            Environment = "production"
          }
          entity_tags_match = "all"
          on_problem_close = false
        }
      }
    }
  }
}

tasks {
  task {
    name      = "send_email"
    description = "Send email notification"
    action     = "dynatrace.email:email-action"
    active     = true
    input = jsonencode({
      to      = ["team@example.com"]
      subject = "Dynatrace Problem: {{event()['event.name']}}"
      body    = "Problem detected.\nName: {{event()
['event.name']}}\nSeverity: {{event()['event.category']}}"
    })
    position {
      x = 0
      y = 1
    }
  }
}
```

Grail Dashboard (Document)

```
resource "dynatrace_document" "team_dashboard" {
  type    = "dashboard"
  name    = "Production Overview"
  private = false
  content = jsonencode({
    version = 13
    variables = []
    tiles = {
      "tile-1" = {
        type    = "data"
        title   = "Error Rate"
        query   = "timeseries avg(dt.service.request.failure_rate), by:
{dt.entity.service}"
      }
    }
    layouts = {
      "tile-1" = { x = 0, y = 0, w = 12, h = 6 }
    }
  })
}
```

Grail Notebook (Document)

```
resource "dynatrace_document" "runbook" {
  type    = "notebook"
  name    = "Incident Runbook"
  private = false
  content = jsonencode({
    version = "3"
    sections = [
      {
        id      = "section-1"
        type    = "markdown"
        title   = "Investigation Steps"
        content = "## Step 1: Check Error Rate\nRun the query below to\nidentify affected services."
      },
      {
        id      = "section-2"
        type    = "dql"
        title   = "Error Rate by Service"
        content = "timeseries avg(dt.service.request.failure_rate), by:\n{dt.entity.service}"
      }
    ]
  })
}
```

Segment (Gen3 Management Zone Replacement)

```
resource "dynatrace_segment" "production_services" {
  name          = "Production Services"
  description    = "All services in the production environment"
  is_public     = true
  include_filter = "fetch dt.entity.service | filter tags =\n\"Environment:production\""
}
```


Maintenance Window

```
resource "dynatrace_maintenance" "weekly_patch" {
  enabled = true
  name    = "Weekly Patch Window"
  type    = "PLANNED"
  suppression = "DONT_DETECT_PROBLEMS"

  schedule {
    type = "WEEKLY"
    weekly_recurrence {
      day_of_week    = "SUNDAY"
      time_window {
        start_time = "02:00"
        end_time   = "04:00"
      }
      recurrence_range {
        start_date = "2026-01-01"
      }
    }
  }
}
```

5. State Management

Understanding Terraform State

Terraform tracks resources in a state file (`terraform.tfstate`):

Concept	Description
State	JSON file mapping config to real resources
Plan	Comparison of state vs. desired config
Apply	Execute changes to reach desired state
Drift	Difference between state and actual

Remote State Storage

For team collaboration, use remote state:

```
terraform {
  backend "s3" {
    bucket = "my-terraform-state"
    key    = "dynatrace/production/terraform.tfstate"
    region = "us-east-1"
  }
}
```

Or using Terraform Cloud:

```
terraform {
  cloud {
    organization = "my-org"
    workspaces {
      name = "dynatrace-production"
    }
  }
}
```

Common State Commands

```
# Preview changes
terraform plan

# Apply changes
terraform apply

# Show current state
terraform show

# List resources in state
terraform state list

# Import existing resource
terraform import dynatrace_management_zone_v2.production "<object-id>"

# Remove resource from state (without deleting)
terraform state rm dynatrace_management_zone_v2.production
```

Importing Existing Resources

To manage existing configurations:

1. Write the resource block in HCL
2. Find the object ID in Dynatrace
3. Import into state

```
terraform import dynatrace_management_zone_v2.production  
"vu9U3hXa3q0AAAABABhidWlsdGlu0m1hbmFnZW1lbmQtem9uZXMAbNlbnFudAAGdGVuYW50ABp2  
dTlVM2hYYTNxMERUVF9fUHJvZHVjdGlvbr7vFJ4"
```

6. Advanced Patterns

Modules for Reusability

Create reusable modules:

modules/environment/main.tf:

```

variable "environment_name" {
  type = string
}

variable "environment_tag" {
  type = string
}

resource "dynatrace_management_zone_v2" "zone" {
  name = var.environment_name

  rules {
    type      = "SERVICE"
    enabled   = true

    conditions {
      condition {
        key {
          type      = "STATIC"
          attribute = "SERVICE_TAGS"
        }
        tag {
          operator = "EQUALS"
          negate   = false
          value {
            context = "CONTEXTLESS"
            key     = "environment"
            value   = var.environment_tag
          }
        }
      }
    }
  }
}

output "management_zone_id" {
  value = dynatrace_management_zone_v2.zone.id
}

```

Use the module:

```
module "production" {
  source          = "../modules/environment"
  environment_name = "Production"
  environment_tag  = "production"
}

module "staging" {
  source          = "../modules/environment"
  environment_name = "Staging"
  environment_tag  = "staging"
}
```

Workspaces for Environments

Use workspaces to manage multiple environments:

```
# Create workspaces
terraform workspace new development
terraform workspace new staging
terraform workspace new production

# Switch workspace
terraform workspace select production

# List workspaces
terraform workspace list
```

Use workspace in config:

```
locals {
  environment = terraform.workspace

  config = {
    development = {
      alert_delay = 30
    }
    staging = {
      alert_delay = 15
    }
    production = {
      alert_delay = 5
    }
  }
}

resource "dynatrace_alerting" "alerts" {
  name = "${local.environment} Alerting"

  rules {
    delay_in_minutes = local.config[local.environment].alert_delay
    severity_level   = "AVAILABILITY"
  }
}
```

Modules with Governance Inputs

For multi-team environments, modules can enforce governance through input validation, mandatory tags, and scoping:

```
# modules/alerting-profile/variables.tf

variable "team_name" {
  type      = string
  description = "Name of the team owning this alerting profile"
  validation {
    condition     = can(regex("^[a-z][a-z0-9-]{2,20}$", var.team_name))
    error_message = "Team name must be lowercase alphanumeric with hyphens, 3-21 characters."
  }
}

variable "management_zone_name" {
  type      = string
  description = "Management zone to scope this profile to (required)"
  validation {
    condition     = length(var.management_zone_name) > 0
    error_message = "Management zone name is required."
  }
}

variable "environment" {
  type      = string
  description = "Target environment"
  validation {
    condition     = contains(["dev", "staging", "prod"], var.environment)
    error_message = "Environment must be one of: dev, staging, prod."
  }
}

variable "error_delay_minutes" {
  type      = number
  default    = 0
  validation {
    condition     = contains([0, 5, 10, 15, 30], var.error_delay_minutes)
    error_message = "Error delay must be one of: 0, 5, 10, 15, 30 minutes."
  }
}
```

This pattern ensures teams cannot create unscoped or untagged resources — governance is built into the module itself.

IAM Policy Management via Terraform

The Dynatrace Terraform provider can manage IAM policies, groups, and bindings. This enables **schema-level access restrictions** — something API tokens cannot do.

Important: Managing IAM resources requires OAuth client credentials with `DT_ACCOUNT_ID` . API tokens cannot manage IAM.

```
# Create a policy that restricts a team to specific settings schemas
resource "dynatrace_iam_policy" "team_settings" {
  name          = "Payments Team - Settings Access"
  environment    = var.dynatrace_environment_id
  statement_query = &lt;&lt;-EOT
    ALLOW settings:objects:read, settings:objects:write
    WHERE settings:schemaId IN (
      "builtin:alerting.profile",
      "builtin:problem.notifications",
      "builtin:maintenance-window"
    );
  EOT
}

# Create an IAM group for the team
resource "dynatrace_iam_group" "payments_team" {
  name          = "Payments Team"
  description    = "IAM group for the Payments team"
}

# Bind the policy to the group
resource "dynatrace_iam_policy_bindings_v2" "payments_binding" {
  group        = dynatrace_iam_group.payments_team.id
  policy_id     = dynatrace_iam_policy.team_settings.id
  environment   = var.dynatrace_environment_id
}
```

IAM policies support multiple condition operators:

Operator	Example	Use Case
<code>IN</code>	<code>settings:schemaId IN ("builtin:alerting.profile", ...)</code>	Explicit list
<code>startsWith</code>	<code>settings:schemaId startsWith "builtin:alerting"</code>	Schema family
<code>contains</code>	<code>settings:schemaId contains "custom"</code>	Substring match
<code>=</code>	<code>storage:bucket.name = "team_logs"</code>	Data isolation

Key insight: The API token scope `settings.write` grants access to ALL schemas. IAM policies with `WHERE settings:schemaId` clauses (using the IAM action `settings:objects:write`) are the only way to restrict schema access at the platform level.

Best Practices

Practice	Description
Remote state	Never use local state for teams
State locking	Enable to prevent concurrent changes
Modular design	Use modules for reusable patterns
Version pinning	Pin provider versions
Plan before apply	Always review plan output
Meaningful names	Resource names should be descriptive

Common Issues

Issue	Cause	Solution
"Resource already exists"	State out of sync	Import the resource
"Provider error"	Invalid token	Check credentials
"Drift detected"	Manual changes	Re-apply or update state
Slow plan/apply	Many resources	Use targets or modules

7. Next Steps

Terraform Export Utility

To export existing Dynatrace configuration to Terraform:

```
# Install terraform-provider-dynatrace-export
# See: https://github.com/dynatrace-oss/terraform-provider-dynatrace

# Export all configurations
terraform-provider-dynatrace-export \
  -e https://{tenant}.live.dynatrace.com \
  -t {api-token} \
  -o ./exported-config
```

Continue the Series

Next Notebook	Focus
AUTOM-05: Dynatrace Workflows	Event-driven automation

Additional Resources

- [Terraform Provider Documentation](#)
- [Provider GitHub Repository](#)
- [Terraform Style Guide](#)

Summary

In this notebook, you learned:

- How to configure the Dynatrace Terraform provider
- Creating resources like management zones, auto-tags, and SLOs
- State management and drift detection
- Advanced patterns with modules and workspaces

Key Takeaway: Terraform excels when you need state management, drift detection, or integration with other infrastructure. For Dynatrace-only config management, Monaco is often simpler.

Continue to AUTOM-05: Dynatrace Workflows to learn event-driven automation.

This notebook was AI-generated from community-submitted and publicly available sources. This notebook series is not officially supported by Dynatrace. Always verify information against official Dynatrace documentation.